Maths Exploring at the end of the year-04

Progressive exercises

E1 · Direct application. Determine $\lim_{x\to+\infty} \dfrac{5x^3 - x}{2x^3 + 7}$ justifying.

E2 · Application and justification. Let f the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits in 2^- in 2^+, in $-\infty$ and in $+\infty$.

E3 · Method to be organised. Determine $\lim\limits_{x\to1} \dfrac{x^2 - 1}{x - 1}$ specifying why the simplification used is valid in the vicinity of 1.

E4 · Choice of strategy. Determine $\lim\limits_{x\to+\infty} x^3 e^{-2x}$. Cite the comparative growth result used.

E5 · Synthesis or demonstration. Determine $\lim\limits_{x\to-\infty} \left(\sqrt{x^2+1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.